



College of Engineering
UNIVERSITY OF WISCONSIN-MADISON

Syllabus

Course Information

Course Subject, Number and Title

E C E 553 - Testing and Testable Design of Digital Systems

Credits and Contact Hours Per Week

3 credits, 3 contact hours per week

Credit Hour Explanation

This course follows the Traditional Carnegie Definition. The class meets for two 75-minute class periods each week over the fall semester and carries the expectation that students will work on course learning activities (watching videos, reading, problem sets, studying, working on the project, etc.) for about 3 hours out of classroom for every class period.

Course Description

Faults and fault modeling, test equipment, test generation for combinational and sequential circuits, fault simulation, memory and microprocessor testing, design for testability, built-in self-test techniques, and fault location.

Requisites

E C E/COMP SCI 352 and COMP SCI 300, graduate/professional standing, or member of Engineering Guest Students

Course Designations/Attributes

General education

Learning Outcomes

Course Learning Outcomes

Upon successful completion of this course, undergraduate and graduate students will be able to

...

1. Identify factors that impact economics of testing of integrated circuits.
2. Model and simulate faults that can occur in integrated circuits
3. Apply state-of-the-art test generation algorithms
4. Utilize design for testability and built-in self-test techniques

Additionally, graduate students will be able to . . .

5. Generate tests to detect path delay faults

ABET Outcomes

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics

6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions

Meeting and Instructor Information

Meeting Time and Location

TR 9:30-10:45am; 3345 Engineering Hall

Discussion and/or Lab Section Information

N/A

Instructional Modality

In-person

Instructor Title and Name

Professor Azadeh Davoodi

Instructor Contact Information

adavoodi@wisc.edu

Day/Time/Location of Office Hours

Wednesdays 12:00 PM - 1:00 PM,

Thursdays 12:00 PM – 1:00 PM at **3625 EH**

Please check the Office Hours page on Canvas before attending. A few office hours may be canceled which will be announced online.

Teaching Assistant Name(s)

Navid Nader Tehrani

Teaching Assistant Contact Information

nadertehrani@wisc.edu

Day/Time/Location of TA Office Hours

Mondays 4:00 – 5:00 PM at B555,
Tuesdays 4:30 – 5:30 PM (virtual on <https://uwmadison.zoom.us/j/2270343841>)

Overview

Brief List of Topics

- Test Economics
- Fault Modeling and Fault Simulation
- Test Generation Algorithms
- Functional testing
- Memory testing
- Design for Testability including boundary scan
- Built-In Self-Test
- Test compression

Materials

Link to Canvas Site

<https://canvas.wisc.edu/courses/516790>

Other Digital Instructional Tools (if applicable)

Video lectures corresponding to a small portion of the slides (often supplementary material) will be posted on the Canvas page.

Required Textbook(s), Software and Other Materials

(WWW) L.T. Wang, C.W. Wu, and X. Wen, “VLSI Test Principles and Architectures”, Morgan Kaufmann, 2006.

(BA) Michael L. Bushnell and Vishwani D. Agrawal, Essentials of Electronic Testing for Digital, Memory, and Mixed-Signal VLSI Circuits, Kluwer Academic Publishers, Boston, 2000.

The above are not required. Electronic versions are linked from the Canvas page and may be borrowed from the library.

Coursework and Grading

Assignments/Exams/Other Graded Work

HOMEWORKS: There will be about 9 homework provided as sample problems along with their solutions. You are expected to first work on each homework before checking the solution. Homework is not graded. Together with the slides, they will serve as the basis for the quizzes.

QUIZZES: There will be about 9 in-class quizzes. For each student, the quiz grade is calculated by dropping their lowest quiz grade and averaging the rest. This is to account for unforeseen circumstance such as missing a quiz due to illness, etc. Therefore there won't be any make-up quiz.

PROJECT: The student project will involve use of the workstations in CAE. You are expected to know use of the Unix operating system, editors, and applications software. Students are allowed to use generative AI tools to improve their project and to debug their code. Students who use AI for the course project are required to cite and fully explain this usage (tool used, provided prompts, agents, etc.) as a separate document which should be submitted with the project material. All students are required be able to explain their project including any AI-generated content, if requested by the instructor.

EXAMS: There will be one midterm and one final exam. The midterm is scheduled on Wednesday 11/04/2026 from 5:45-7:15pm. There will be an alternate midterm exam scheduled on Monday 11/09/2026 from 10:00am-noon. Taking an alternate exam is only possible with a valid excuse due to an unexpected circumstance, and only if accepted by the instructor. Scheduling travel during the midterm date is not a valid excuse. Final exam is on Saturday 12/12/2026 from 2:45-4:45pm. All exams will be closed-book closed-notes. Electronic/smart devices including calculators are not allowed. Use of one 8 ½" by 11" cheat sheet (single-sided) is allowed.

Exam Proctoring

Instructors have the authority to decide whether to proctor their tests, quizzes or other course assessments. Failure to accept the proctoring assigned will result in zero on assignments.

Grading

- Written Homeworks (WH): 0% (not graded)
- Applied Homeworks (AH): 6% (AH1:3, AH2:3)
- Quizzes (Q): 22% (for 8 highest-grade quizzes)
- Project (P): 20% (P1:5, P2:15)
- Midterm (M): 25%
- Final Exam (F): 25%
- Participation: 2%

* Graduate students will have an additional component to their project.

Your final grade will be assigned as a letter grade according to the following scale and may be curved up to the benefit of students.

A	90-100%
B	80-89%
C	70-79%
D	60-69%
F	0-59%

Tentative Schedule

Week	Topic	Activities
1 R: 9/3 T: 9/8	Introduction to Testing, Types of Testing, Test Quality, Test Economics	Reading: WWW: 1.1, 1.2 - BA: 1.4, 2.1, 3.2, 3.3 WH1 released
2 R: 9/10 T: 9/15	Logic Simulation, Fault Modeling	Reading: WWW: 3.2, 3.3 - BA: 5.1-5.4 Quiz 1 on 9/15 (based on HW1) HH2 released
3 R: 9/17 T: 9/22	Fault Modeling, Fault Collapsing	Reading: BA 4.5 Quiz 2 on 9/22 (based on HW2) WH3 released
4 R: 9/24 T: 9/29	Fault Collapsing , Fault Simulation	Reading: WWW: 3.4 - BA 5.5, 5.6
5 R: 10/1 T: 10/6	Testability Measures	Reading: WWW 2.2 - BA 6.1 WH4 released Quiz 3 on 10/1 (based on HW3)
6 R: 10/8 T: 10/13	Combinational ATPG	Reading: WWW 4.1-4.4 AH1 released Quiz 4 on 10/8 (based on HW4)
7 R: 10/15 T: 10/20	Combinational ATPG	Reading: BA 7.5 AH1 due 10/15 WH5 released
8 R: 10/22 T: 10/27	Sequential ATPG	Reading: WWW 4.5 - BA 8.2, 8.3 WH6 released Quiz 5 on 10/22 (based on HW5)
9 R: 10/29 T: 11/3	Transition and Path Delay Testing	Reading: WWW 2.3-2.5 - BA 14 AH2+3 released Quiz 6 on 11/3 (based on HW6)
10 R: 11/5 T: 11/10	Design for Testability: Scan Chains	Reading: WWW 10.2 - BA 16 Midterm (11/4) WH7 released
11 R: 11/12 T: 11/17	Design for Testability: JTAG	Reading: WWW 5.3-5.5 - BA 15 WH8 released AH2+3 due 11/11 P1 released Quiz 7 on 11/17 (based on HW7)
12 R: 11/19 T: 11/24	Built-in Self-Test	Reading: WWW 8.2 - BA 9.5-9.7 WH9 released Quiz 8 on 11/24 (based on HW8) P1 due 11/19 ; P2 released
T: 11/26	Thanksgiving Holiday	
13 T: 12/1 R: 12/3	Test Compression, Memory Test	Reading: WWW 6 Quiz 9 on 12/3 (based on HW9) Project part 2 due 12/3
14 T: 12/8	Functional Test	Video to watch Final on 12/12/2026 , 2:45-4:45pm

Policies

Academic Integrity

By virtue of enrollment, each student agrees to uphold the high academic standards of the University of Wisconsin–Madison; academic misconduct is behavior that negatively impacts the integrity of the institution. Cheating, fabrication, plagiarism, unauthorized collaboration and helping others commit these previously listed acts are examples of misconduct which may result in disciplinary action. Examples of [disciplinary sanctions](#) include, but are not limited to, failure on the assignment/course, written reprimand, disciplinary probation, suspension or expulsion.

Use of Artificial Intelligence (AI) Tools

You are welcome to use artificial intelligence (AI) tools and applications in this class as they support the learning objectives of this course. Please be aware you are responsible for the information you submit based on an AI query (i.e. ensure instructor has allowed you to publicly post course content such as assignment or assessment prompts and that the AI generated results do not contain misinformation or unethical content). Your use of AI tools must be documented and cited to conform to this course's expectations.

Accommodations for Students with Disabilities

The University of Wisconsin–Madison supports the right of all enrolled students to a full and equal educational opportunity. The Americans with Disabilities Act (ADA), Wisconsin State Statute (36.12) and UW–Madison policy ([UW-855](#)) require the university to provide reasonable accommodations to students with disabilities to access and participate in its academic programs and educational services. Faculty and students share responsibility in the accommodation process. Students are expected to inform faculty of their need for instructional accommodations during the beginning of the semester, or as soon as possible after being approved for accommodations. Faculty will work either directly with the student or in coordination with the [McBurney Disability Resource Center](#) to provide reasonable instructional and course-related accommodations. Disability information, including instructional accommodations as part of a student's educational record, is confidential and protected under FERPA.

Engineering for All

We cultivate and uphold a safe, welcoming, and supportive environment for all, operating with the highest standards of integrity, collegiality, and ethics and recognizing the contributions of every member of our college community. Our faculty integrate innovative, multidisciplinary research into the classroom, leveraging evidence-based teaching methods to inspire and prepare our students. Through our research, teaching and service, we pursue answers to societal needs and challenges.

Academic Calendar and Religious Observances

View the full [academic calendar](#) in addition to information about religious and election day observances. Students are responsible for notifying instructors within the first two weeks of classes about any need for flexibility due to [religious observances](#).

Establishment of the academic calendar for the University of Wisconsin–Madison falls within the authority of the faculty as set forth in [Faculty Policies and Procedures](#). Construction of the academic calendar is subject to various rules and laws prescribed by the Board of Regents, the Faculty Senate, State of Wisconsin and the federal government. Find [additional dates and deadlines for students](#) on the Office of the Registrar website.

Mental Health and Well-Being

As a student you may experience a range of issues that can cause barriers to learning. These might include strained relationships, anxiety, high levels of stress, alcohol/drug problems, feeling down, or loss of motivation. University Health Services can help with these or other issues you may experience. Help is always available. You can learn about free, confidential mental health services available to you; call 608-265-6600 (option 2) or visit uhs.wisc.edu. College of Engineering students can also directly access short-term mental health support from the engineering embedded mental health provider. Visit the [Engineering Wellness](#) page for more information.

Privacy of Student Records & Use of Audio Recorded Lectures

Lecture materials and recordings for this course are protected intellectual property at UW–Madison. Students enrolled in this course may use the materials and recordings for their personal use related to participation in the course. Students may also take notes solely for their personal use. If a lecture is not already recorded, students are not authorized to record lectures without permission unless they are considered by the university to be a qualified student with a disability who has an approved accommodation that includes recording. [Regent Policy Document 4-1] Students may not copy or have lecture materials and recordings outside of class, including posting on internet sites or selling to commercial entities, with the exception of sharing copies of personal notes as a notetaker through the McBurney Disability Resource Center. Students are otherwise prohibited from providing or selling their personal notes to anyone else or being paid for taking notes by any person or commercial firm without the instructor's express written permission. Unauthorized use of these copyrighted lecture materials and recordings constitutes copyright infringement and may be addressed under the university's policies, UWS Chapters 14 and 17, governing student academic and non-academic misconduct. View [more information about FERPA](#).

Teaching & Learning Data Transparency

The privacy and security of faculty, staff and students' personal information is a top priority for UW–Madison. The university carefully reviews and vets all campus-supported digital tools used for teaching and learning, including those that support [data empowered educational practices](#) and proctoring. View more information about [teaching and learning data transparency](#) at UW–Madison.

Course Evaluations

Students at the University of Wisconsin–Madison have the opportunity to evaluate their learning experiences and the courses they are enrolled in through course evaluations. Many instructors use a digital course evaluation tool to collect feedback from students. Student participation is an integral component of course development, and confidential feedback is important. UW–Madison strongly encourages student participation in course evaluations.

Students' Rules, Rights & Responsibilities

View more information about [student rules, rights and responsibilities](#) such as student privacy rights, sharing of academic record information, academic integrity and grievances. Find additional College of Engineering information about [reporting grievances and complaints](#).